

Chun-Kai Yang

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in Chun-Kai Yang

🔗 YCK1130

📁 Profolio

Education

University of Michigan

Jun 2028 (Est.)

Master of Science in Robotics

- **Coursework:** Robotic Systems Laboratory, Mobile Manipulation

National Taiwan University

Jan 2026

Bachelor of Science in Engineering - Electrical Engineering

CGPA: 3.97/4.0

- **Coursework:** Robot Sensing and Control(A+), Reinforcement Learning(A+), Deep Learning in Computer Vision(A), Machine Learning(A+), Introduction to IoTs(A), Embedding System(A+)
- 2021-Fall Dean's List Award (Top 5% in Department)

Work Experience

Lasertec Taiwan Inc.

Taipei, Taiwan

Data Scientist Intern

May 2025 - Sep 2025

- Designed a Deep-Learning-based defocus image correction for photolithography mask anomaly detection, improving precision 15% via a custom data synthesis and evaluation pipeline
- Accelerated inference by 23.4x via multi-threaded parallel processing and caching of repeated inputs, cutting analysis time from 2-3 days to under 2 hours and enabling full-batch analysis across all production line data instead of sampling
- Designed metric analysis across 30+ metrics to uncover correlations between anomaly indicators, enabling rule-based metric combinations that filtered out extreme outlier anomalies

Instill AI

London, UK

AI Engineering Intern

Jun 2024 - Aug 2024

- Integrated GitHub, Jira, and Asana's APIs for a low-code ETL platform, implementing 20+ CRUD and webhook functionalities that let users integrate services without writing code, covering 80% of common use cases
- Developed a configuration-driven testing framework with shared mock server logic, letting engineers write only test case inputs/outputs without rewriting mock server test logic for each service

Research Experience

Robot Learning Lab, NTUEE

Taipei, Taiwan

Undergraduate Researcher, Advisor: Prof. Shao-Hua Sun

Jan 2023 - Jan 2026

- Designed a diffusion-based imitation learning framework [2] and an optical-flow-based skill learning framework [1], boosting performance 66% on high-DoF manipulation and locomotion tasks, and data efficiency 2x
- Deployed the framework on ALOHA, diagnosing failures including sensor reading anomalies and wiring disconnections; designed a control safeguard algorithm to constrain joint angles and prevent hardware damage
- Evaluated VLA and video generation models for synthetic data generation, identifying that action-space misalignment with the target embodiment was the key bottleneck to usable motion generation

Publications

- [1] **Learning Skills from Action-Free Videos** [Paper]

ICML WM Workshop, 2025

Hung-Chieh Fang*, Kuo-Han Hung*, Chu-Rong Chen, Po-Jung Chou, **Chun-Kai Yang**, et al.

- [2] **Diffusion Imitation from Observation** [Paper]

NeurIPS, 2024

Bo-Ruei Huang, **Chun-Kai Yang**, Chun-Mao Lai, Dai-Jie Wu, Shao-Hua Sun

Academic Projects

Multimodal Perception & Comprehension in Autonomous Driving [Code][Poster]

Nov 2024 - Dec 2024

- Developed a 2-stage RAG-enhanced LLaVA system for autonomous driving perception, using segmentation and depth features to improve spatial understanding, increasing evaluation score by 21% over baseline LLaVA
- Optimized multi-GPU and communication efficiency between devices, decreasing training time by 5x

Edge-AI Proximity Warning System for Power Infrastructure [Code]

Oct 2024 - Dec 2024

- Developed an end-to-end TinyML pipeline, including COCO-format annotation, Swift YOLO training on SSCMA, and int8 quantization from PyTorch to TensorFlow Lite, enabling on-device crane-to-tower proximity detection on ESP32S3
- Integrated two GPIO-alarmed camera nodes with a Flask server over Wi-Fi, allowing operators to redraw warning lines on the live stream and dispatching timestamped Line notifications upon violation

Reinforcement Logic Optimization for a General Cost Function [Code][Report] May 2024 - Jun 2024

- Developed a reinforcement learning framework for logic synthesis with Yosys ABC, designing effective state- and action-space and a reward function under limited information provided by tools
- Outperformed baselines, including Greedy and Simulated Annealing, across all netlist and cost estimator

Collaborative Review for Intelligent Code Analysis using LLMs [Code][Report] May 2024 - Jun 2024

- Developed a multi-agent framework to identify vulnerabilities and discrepancies for code review
- Received higher human preference over CodeQL, with higher efficiency and coverage

Technical Projects

Preliminary Diagnostic and Disease Monitoring Medical Service [Code] Sep 2024

Taipei Codefest (Hackathon), Taipei City Government

- Developed a full-stack application that visualizes yearly disease trends using government open data and provides location-based clinic recommendations
- Integrated Large Language Models (LLMs) to deliver personalized guidance and redirected users to the most suitable hospital websites for treatment

NTUEE Light Dance [Code][Demo] Oct 2022 - Mar 2024

- Designed data storage structure and serialization methods for LED control system, reducing storage usage by 30% while maintaining fast access and scalability for large-scale performances
- Collaborated in a 50+ member student-initiated project, improving maintainability and enabling convenient feature extension by refactoring lagacy codebase into modular components
- Developed main algorithms and CLI tools, achieving 2.5x frame rate with parallel processing

Leadership Experience

NTUEE Student Association Sep 2023 - Jun 2024

Director of Information Department

- Directed a 10+ members team to develop and maintain platforms for student association activities; created React-based mini-games that engaged 200+ participants
- Organized workshops on web design, Linux, and Git/GitHub, attracting 80+ attendees

NTUEE Night Sep 2023 - Apr 2024

General Coordinator

- Directed 100+ performers and staff across multiple teams, managing task allocation, scheduling, and on-site execution for seamless event operations
- Managed venue preparation, program flow, vendor negotiation, and final approval of event deliverables

Nine-Department Intercollegiate Christmas Ball Sep 2022 - Dec 2022

Deputy Director of General Affairs

- Coordinated 50+ staff through training, task allocation, and scheduling
- Managed venue layout, program flow, and on-site execution with real-time contingency handling

Skills

Programming: C/C++, Python, Golang, CUDA, TypeScript, MongoDB, SQLite, PostgreSQL

Frameworks: PyTorch, TensorFlow, OpenCV, Lightning, ROS2, MuJoCo, ALOHA, NodeRED, React

Tools: Raspberry Pi, Arduino, ESP32, git, Docker, Linux, Fusion 360, AutoCAD

Languages: Mandarin (Native), English (Fluent)